

NEGATIVE NUMBERS — LESS THAN ZERO!

"Numbers That Go the Other Way!"

■ Course	Negative Numbers — Intro
■■ Session	Session 1 of 3
■■ Duration	30 Minutes
■ Depth	Beginner
■ Board	CBSE / ICSE / State Board
■■ Language	English (Indian English)

WHAT IS THIS LESSON ABOUT?

This session introduces learners to **negative numbers** — what they are, where they appear in real Indian life, and how to read and place them on a number line. No computation with negative numbers is required. The focus is entirely on **recognition, comparison, and intuitive understanding** through everyday Indian contexts like mall lift basement floors, winter temperatures in Shimla, and bank balances.

■ LIFT FLOORS B1, B2, B3 =
-1, -2, -3

■■ TEMPERATURES Shimla
-3°C Kargil -8°C

■ BANK BALANCE Owing ■5 =
-■5

LEARNING OBJECTIVES

KNOWLEDGE — What learners will KNOW

LO-K1	State that numbers below zero are called negative numbers.
LO-K2	Recognise that zero is neither positive nor negative — it is the middle point.
LO-K3	Read and say negative numbers correctly (e.g., "negative 3" or "minus 3") when shown -3 .
LO-K4	Identify at least two real-life Indian situations where negative numbers are used.

SKILLS — What learners will be able to DO

LO-S1	Place positive and negative whole numbers (up to -10 and $+10$) on a number line in the correct order.
LO-S2	Say which of two numbers is greater or smaller using the number line (e.g., -2 is greater than -5).
LO-S3	Count backwards through zero from any small positive number into negative numbers.
LO-S4	Match a negative number to its real-life description (e.g., "3 floors below ground" = -3).

STRETCH: LO-S2 is for advanced learners only — comparing and ordering negative numbers.

UNDERSTANDING — Big Ideas learners will UNDERSTAND

LO-U1	Understand that the number line goes both ways — numbers continue beyond zero to the left.
LO-U2	Understand that a larger negative number (e.g., -10) is actually LESS than -1 . (Surprising discovery!)
LO-U3	Understand that negative numbers are not "wrong" numbers — they describe real situations.

ATTITUDE — Disposition Goals

LO-A1	Feel curious and excited about numbers that exist beyond zero.
LO-A2	Build confidence that maths describes real life — even basement floors and cold winters!
LO-A3	Develop the habit of noticing and naming negative number situations in daily Indian life.

Bloom's Taxonomy Mapping

NEGATIVE NUMBERS — INTRO

Planrs Academy · Session 1 of 3 · 30 Minutes · Beginner

PLANRS ACADEMY

"Learning That Feels Like Discovery"

PRIMARY	Remembering + Understanding
SECONDARY	Applying (matching, placing on number line)
STRETCH	Analysing (comparing and ordering negative numbers)

KEY CONCEPTS

CONCEPT 1 — WHAT ARE NEGATIVE NUMBERS?

Simple Definition: Negative numbers are numbers that are LESS than zero. They are written with a minus sign in front — for example, -3 .

Child-Friendly: "You know how we count 1, 2, 3 going UP? Well, we can also count 0, -1 , -2 , -3 going DOWN! These are called **NEGATIVE** numbers."

Notation introduced:

- Written form: -1 , -2 , -3 (with the minus sign close to the number)
- Spoken form: "negative one", "minus one", "negative two"
- Do NOT introduce fraction or decimal negatives at this level.

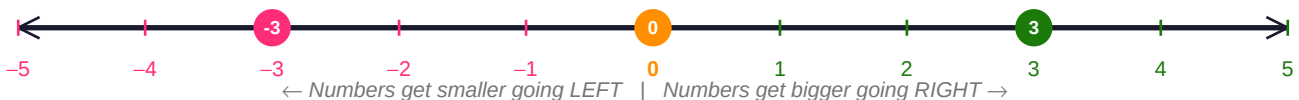
Indian Real-Life Examples:

■ Lift / Elevator	Big malls in Mumbai, Bengaluru, Delhi have floors B1, B2, B3. B1 = -1 , B2 = -2 , B3 = -3 — basements are BELOW the ground floor!
■■ Temperature	In Shimla or Kashmir in January, temperature can be -3°C or -5°C . "Below zero cold!" — a real negative number on the thermometer.
■ Cricket	When India needs 10 more runs to win and scores only 5, the run deficit is -5 runs. The team fell short of their target.
■ Bank / Money	Aarav spent ■50 but only had ■30. He owes ■20. His balance is $-\text{■}20$ — he is "in minus"!

CONCEPT 2 — THE NUMBER LINE GOES BOTH WAYS

Key Idea: The number line does NOT start at zero. Zero is the **MIDDLE** point. To the right are positive numbers. To the **LEFT** are negative numbers.

Visual — The Number Line:



Indian Examples:

- A thermometer in Delhi: June = $+42^{\circ}\text{C}$, Kargil January = -15°C . Goes both ways!
- A lift: Basement 3 (-3), Basement 2 (-2), Basement 1 (-1), Ground (0), Floor 1 ($+1$)...
- If Rohan walks 4 steps forward → $+4$. Then 6 steps back → goes through 0 → lands at -2 !

Teaching Note: Always draw the number line with zero clearly in the **CENTRE**. Never start the number line at zero — this reinforces the misconception that zero is the smallest number.

■ KEY CONCEPTS (CONTINUED)

CONCEPT 3 — ZERO IS THE BOUNDARY

Key Idea: Zero is NEITHER positive NOR negative. It is the point where positive and negative numbers "meet." It means "nothing" — no temperature difference from freezing point, no money in the account, ground floor.

- Ground floor of a mall = Floor 0 (neither up nor down)
- A cricket score of 0 — exactly zero runs, not negative!
- 0°C — the point where water just starts to freeze (not yet negative!)

Teaching Note: Many children assume zero is "the smallest number." Correct this gently by asking: "What is one step BELOW zero on our thermometer?" Let them discover -1 themselves!

CONCEPT 4 — READING AND WRITING NEGATIVE NUMBERS

Key Idea: The minus sign ($-$) in front of a number tells us it is NEGATIVE. -4 is read as "negative four" or "minus four." **It is NOT "take away four" at this stage!**

NUMBER	READ AS	REAL-LIFE MEANING
-1	"negative one" or "minus one"	Basement floor 1 of a mall
-3	"negative three" or "minus three"	3°C below zero in Shimla
-5	"negative five" or "minus five"	Owing ₹5 to a friend
-8	"negative eight" or "minus eight"	8°C below zero in Kargil at night

CONCEPT 5 — ORDERING NEGATIVE NUMBERS (STRETCH — Advanced Learners Only)

Surprising Fact: On the number line, numbers always get BIGGER as you go RIGHT. So -1 is BIGGER than -5 , even though 5 "looks bigger." This is the "surprising discovery" about negative numbers!

"Imagine it's winter. -1°C is almost warm (just one step below freezing). But -10°C is VERY cold (ten steps below freezing). So -1 is a BIGGER (warmer) temperature than -10 . Surprising, right?"

Important: Introduce this ONLY for advanced learners, and ONLY using the number line or thermometer visual. Do NOT teach ordering rules as abstract facts.

SESSION OUTLINE — 30 MINUTES

Phase	Title	Duration
1	Warm-Up & Hook	4 min
2	Introducing Negative Numbers	5 min
3	Indian Real-Life Connections	4 min
4	Guided Practice: Number Line	6 min
5	Independent Practice: Matching	5 min
6	Stretch Challenge (Advanced Only)	3 min
7	Wrap-Up, Vocabulary & Exit Ticket	3 min

1

WARM-UP & HOOK — "What Comes Before Zero?"

Duration: 4 Minutes

Activity: "What Comes Before Zero?" — Quick-fire counting chain

Teacher counts backwards aloud: **5, 4, 3, 2, 1, 0...** Pause dramatically at zero and ask: "*What comes NEXT?*" Let students guess freely. Accept ALL answers. Build suspense!

Anchor Visual:

Show a picture of a mall lift panel with buttons: B2, B1, G, 1, 2, 3. Ask: "*Have you ever been in a basement? What floor was it?*" Connect: "*That basement is BELOW zero — a negative floor!*"

Tone: Playful, mysterious, discovery-first. **DO NOT** explain negative numbers yet — just spark curiosity!

2

INTRODUCING NEGATIVE NUMBERS

Duration: 5 Minutes

Activity: "The Number Line Goes Both Ways" — visual reveal

Show a number line from -5 to $+5$ with zero clearly in the centre. Introduce the term "**negative number**" formally. Say: "*These numbers on the LEFT of zero are called NEGATIVE numbers. We put a minus sign in front — like this: -3 .*"

Story — Aarav's Mall Adventure:

Aarav's family visits a mall in Bengaluru. They take the lift from the food court (Floor 2) all the way down to the parking (Basement 2). As each floor passes, count together: $2 \rightarrow 1 \rightarrow 0 \rightarrow -1 \rightarrow -2$. Basements are BELOW zero — negative floors!

Concepts Addressed: LO-K1, LO-K2, LO-K3

3

INDIAN REAL-LIFE CONNECTIONS

Duration: 4 Minutes

Activity: "Where Do We See Negative Numbers in India?" — brainstorm

IMAGE PROMPT	NEGATIVE NUMBER HIDDEN INSIDE
■ ■ Thermometer in Shimla in January	-3°C
■ Mall lift panel with B1, B2, B3	$B1 = -1$, $B2 = -2$, $B3 = -3$
■ Bank passbook showing negative balance	$-\text{₹}5$ (you owe ₹5)
■ Cricket scoreboard — run deficit	-5 runs (team fell short)

Discussion Question: "Can YOU think of a place at home or in your city where negative numbers hide?"
— Accept all creative, informal answers!

Concepts Addressed: LO-K4, LO-U3

SESSION OUTLINE (CONTINUED)

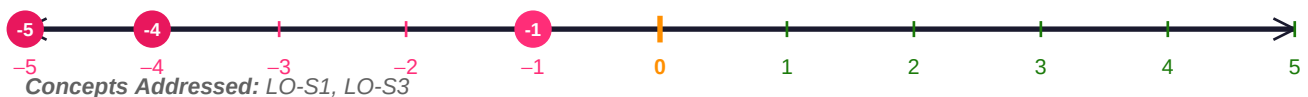
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GUIDED PRACTICE — NUMBER LINE PLACEMENT

Duration: 6 Minutes

Show a number line from -5 to $+5$. Teacher calls out a number; students point to / mark where it belongs. Do 4 questions together.

Q1	Where is -1 on the number line?	One step to the LEFT of zero.
Q2	Where is -4 on the number line?	Four steps to the LEFT of zero — far left!
Q3	Which is further left — -2 or -5 ?	-5 is further left. Further left = smaller!
Q4	Name the number 3 steps to the LEFT of zero.	The answer is -3 .



5

INDEPENDENT PRACTICE — MATCHING ACTIVITY

Duration: 5 Minutes

Students match an Indian real-life situation on the left with the correct negative number on the right. Can be done as worksheet, whiteboard pointing, or digital drag-and-drop.

REAL LIFE SITUATION		NEGATIVE NUMBER
3 floors below ground level	→	-3
Temperature is 5 degrees below zero	→	-5
Priya owes her friend ₹2	→	-2
The lift is at basement floor 1	→	-1

Concepts Addressed: LO-S4, LO-K4

6

STRETCH CHALLENGE — ADVANCED LEARNERS ONLY

Duration: 3 Minutes

Activity: "Which Is Colder?" — comparing two negative temperatures

Delhi winter morning:

-2°C

Just a little below zero

Kargil winter night:

-8°C

Very cold! Much further left on the number line.

Key Point: -2 is to the RIGHT of -8 on the number line. So -2 is GREATER (warmer) and -8 is LESS (colder). "**Further LEFT = smaller = colder.**"

Concepts Addressed: LO-S2, LO-U2 (STRETCH)

Note: Skip this phase if the class needs more time on Phase 4 or 5. This is extension only — do NOT use to assess on-level students.

■ SESSION OUTLINE (CONTINUED)

7

WRAP-UP, VOCABULARY RECAP & EXIT TICKET

Duration: 3 Minutes

Vocabulary to Write on the Board:

NEGATIVE NUMBER

A number LESS than zero, shown with a minus sign (-). Example: -3

ZERO

The middle point — neither positive nor negative. The ground floor!

NUMBER LINE

A line showing numbers going both ways from zero — left is negative, right is positive.

Exit Ticket — 3 Quick Questions:

1	Write ONE negative number.	Any of: -1, -2, -3, -4, -5, etc.
2	Draw a number line and mark -3 on it.	Number line from -5 to +5 with -3 marked correctly (to the left of zero)
3	Tell me one place in India where you might see a negative number.	Accept: mall lift basement, weather app in winter, bank passbook showing

■ Closing Hook: "Next time you see a lift with a Basement button, remember — that's a NEGATIVE number! You're a maths detective now!"

THINGS TO AVOID

CONCEPTUAL ERRORS:

X Teaching that "the minus sign means subtract" — it denotes DIRECTION here, not an operation.
X Saying "you can't have less than nothing" — negative numbers disprove this!
X Introducing rules like "two negatives make a positive" — NOT for this session.
X Using "bigger" to mean greater in magnitude (absolute value) — always use the number line.
X Introducing negative fractions or decimal negatives.
X Any reference to dividing or multiplying negative numbers.

PEDAGOGICAL ERRORS:

X Rushing through the number line visual — it is the primary anchor of the entire session.
X Skipping the real-life Indian connections — they are the conceptual hooks, not decorative.
X Using teacher-talk only for more than 4 consecutive minutes — students must speak/point/count.
X Ending without a vocabulary recap — students must hear and say key terms before closing.
X Treating the stretch activity as core content — it is extension only.
X Starting the number line at 0 — this reinforces the misconception that zero is smallest.

LANGUAGE ERRORS:

X "Easy / Simple / Obviously" — never implies a student should already know something.
X "Math" — always use "Maths" (Indian English convention).
X Dollar signs (\$), miles, or Fahrenheit — use ₹, km, and °C only.
X Western-only names like Tom, Sarah, John — use Indian names (Aarav, Priya, Rohan, Meena).
X Passive voice in student-facing instructions — always active voice.

SUCCESS CRITERIA

STUDENT SUCCESS INDICATORS

80%	■ of students can say "Negative numbers are numbers less than zero" in their own words.
80%	■ of students can correctly place -3 and -1 on a number line when asked.
75%	■ of students can name at least one real-life Indian situation where negative numbers appear.
90%	■ of students attempted all questions in the matching activity (Phase 5).
—	■ Students produced at least 1 spontaneous "That's a negative number!" moment.
75%	■ Exit ticket Q1 and Q2 correct rate $\geq 75\%$.

TEACHER SUCCESS INDICATORS

✓	■ Completed all 7 phases within 30 minutes (± 2 min).
10×	■ Used the term "negative number" at least 10 times during the session.
3×	■ Used the number line visual in at least 3 different explanatory moments.
4×	■ Asked at least 4 open-ended questions (not yes/no).
2×	■ Connected negative numbers to at least 2 Indian real-life examples.
3	■ All vocabulary words on board: NEGATIVE NUMBER / ZERO / NUMBER LINE.

PLATFORM SUCCESS TARGETS

$\geq 85\%$	■ Session completion rate
$\geq 4.2/5$	■ Student satisfaction rating
$\geq 65\%$	■ Parent "Try at Home" card attempted
$\geq 90\%$	■ Teacher feedback "Would use again"
$\geq 2.2/3$	■ Exit ticket average score

MATERIALS LIST

CLASSROOM / LIVE

- Whiteboard or digital display board
- Coloured chalk / markers (red for negative, blue for positive, black for zero)
- Printed number lines (–5 to +5, per student)
- Printed worksheets (1 per student)
- Printed Concept Cards (1 per student)
- Optional: Physical thermometer model
- Optional: Printed lift panel graphic (B3–F3)
- Optional: Coloured counters / stickers

ONLINE / HOME

- Presentation slides (10–18 slides)
- Digital number line (interactive/animated)
- Animated lift panel or thermometer
- Screen-shareable worksheet PDF
- Printable PDF linked in session notes
- Downloadable worksheet PDF
- Downloadable Concept Card PDF
- Parent Communication Card PDF
- Optional: 8–10 question digital quiz

EXTENSION ACTIVITIES

EXT 1	NEGATIVE NUMBER DETECTIVE	Find 3 negative numbers in real life this week (weather app, mall lift, bank app). Share with class.
EXT 2	TEMPERATURE EXPLORER	Track temperatures of 3 Indian cities (north, south, mountain) over one week. Record positive, zero, or negative.
EXT 3	DESIGN YOUR OWN LIFT	Draw a lift panel for a building with 5 floors above and 3 basements. Label each floor with its number.
EXT 4	ORDER THE TEMPERATURES	Arrange: Mumbai +22°C, Delhi +4°C, Shimla –2°C, Kargil –12°C, Bengaluru +18°C from coldest to warmest.
EXT 5	MAKE A QUESTION	Write your OWN word problem using a negative number from something you see at home or in your neighbourhood.

QUICK REFERENCE CARD — FOR CONTENT PRODUCERS

Course	Negative Numbers — Intro Session 1 Beginner 30 minutes
Key Words	Negative Number, Zero, Number Line, Minus Sign
Notation	Written form (–3) and spoken form ("negative three") ONLY. No fractions/decimals.
DO NOT USE	Computation, operations, "integers" (term), fractions, Western-only examples
Number Range	Whole numbers only, within –10 to +10
Indian Refs	Min 4 — Mall Basements, Winter Temperatures, Bank Balance, Cricket Deficit
Names	Aarav, Priya, Rohan, Meena, Kavya, Arjun (rotate)
Tone	Warm, curious, discovery-first, encouraging
CPA Approach	Concrete → Pictorial → Abstract (always in this order)
Review Gate	SME (Primary Maths) + Cultural Context Review required before publication

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